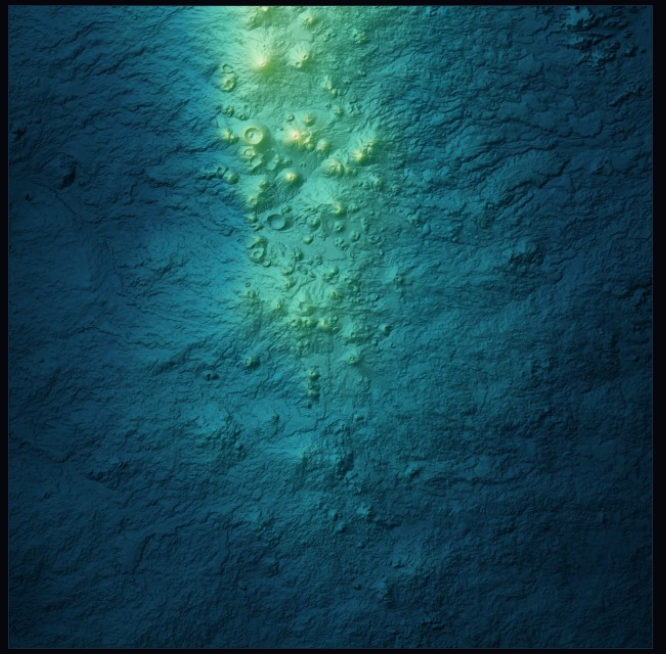
TerraTrace

Earth terrains that answer to the Moon and Mars

Submitted to the NASA Space Apps Challenge 2026 challenge “Identify Earth Locations that Analog the Permanent Moon Base Locations and Mars”. Published as open prior art, built and gated weeks before the event — stated openly rather than dressed up as event work.

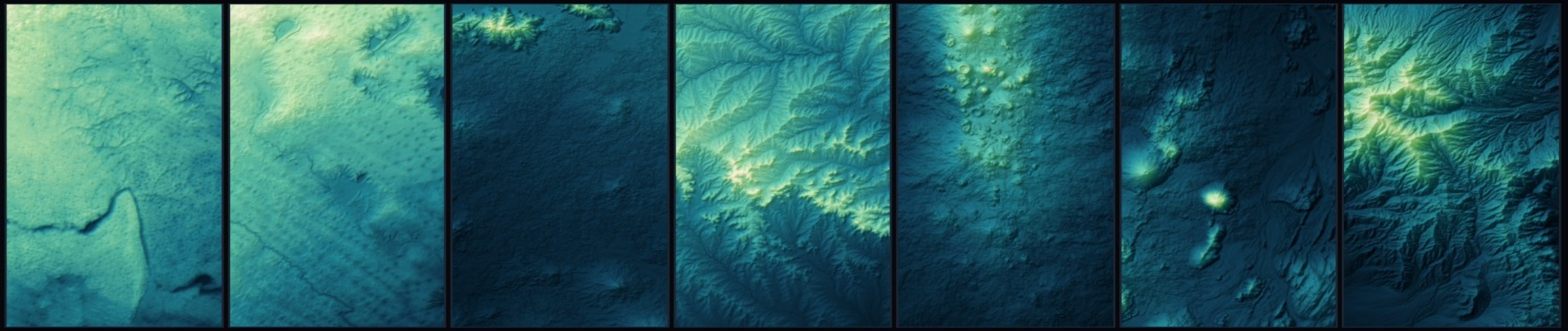
terratrace.fit

`github.com/pal-123456789/terratrace`



Harrat Khaybar, Saudi Arabia · real Copernicus GLO-30 terrain

Seven Earth candidates, one light direction and one vertical scale across every panel — nothing contrast-stretched, so flat terrain reads as flat.



THE QUESTION

Which Earth terrains are morphologically comparable — and at what scale?

One narrow, measurable question. TerraTrace answers it with **measured distance between real elevation models**, and with the scale at which the comparison is honest. It does not answer “where should we land”, and it says so everywhere it could be misread.

What it does

- Reprojects every terrain onto its own body's metric grid.
- Aggregates to a common support, so no unit is compared below its source resolution.
- Ranks by Pareto dominance over declared axes — never a composite score.

What it refuses

- Traversability, autonomy, mission readiness.
- Geological equivalence between two bodies.
- Site access, permitting or ecology.
- Any single “Mars-likeness” score.

Five result states, three of them refusals — including BLOCKED, when a unit cannot be compared at all. A tool that will not say “this is a safe place to land” is more trustworthy for saying so.

WHAT IT IS BUILT ON

Five worlds, one method, no black box

5
planetary bodies: Earth, Mars, the Moon, Mercury, Ceres

40
cited descriptor families (~290 scalar values)

970
offline tests, 0 network required to run them

51
scientific-correctness oracles over the operators

Real data, keyless

- Mars MOLA · Moon LOLA · Mercury MESSENGER · Ceres Dawn.
- Earth: ESA Copernicus GLO-30, read straight from the public AWS COG mosaic — no account, no key.
- 15 catalogued products, each with coverage, licence, vertical accuracy and a per-file SHA-256.

Depth where it is cited, restraint where it is not

- Slope, relief, ruggedness, connectivity, curvature, roughness/Hurst, openness, geomorphons, anisotropy.
- Every descriptor carries a definition, a unit, a window and a citation — and a stated claim boundary.
- A mechanical guard fails the build if any published number stops matching the code.

RESULT 1 · MARS · PROTOCOL V3

It caught its own false analog

An earlier exploratory run favoured “Craters of the Moon, Idaho” — a famous basalt analog. We tested that under a protocol frozen before the data, and **withdrew it ourselves**.

S	SUPPORTED	smoothness tracks the class
V	NOT SUPPORTED	volcanic origin adds nothing
R	SUPPORTED	no candidate reversal under resolution matching

15 / 15 confirmatory cells accounted for. Every resolution-change class is reported, including the two that are zero: 11 unchanged, 0 stabilised, 1 destabilised, 0 reversed, 3 membership changed, 0 error.

Read it honestly

v3 is pre-specified but not fully pre-registered — the rules and config were frozen before the data, the protocol document was committed after the run. We say so on every surface rather than letting a reader discover it.

Within this panel and lens: roughness tracks the class, origin does not.

RESULT 2 · THE MOON · REAL LOLA TERRAIN · EXPLORATORY

The other half of the challenge, run on real lunar data

All **six** published south-polar candidate regions from the NASA Goddard 5 m/pixel LOLA products, screened against the same Earth roster on an **exact common grid**: 500 / 750 / 1000 m cells covering 9 km on every unit, both bodies.

18 / 18

site × support cells where the same single Earth candidate is non-dominated

28 / 30

adversarial perturbations, declared before the run, that leave the ordering untouched

1 of 6

sites with no candidate within 50% on every axis — screened, not matched

The honest shape of it

Ordering is decisive everywhere; closeness at five of six sites. At Shackleton rim the survivor is 70% away on its worst axis — the least-dissimilar member of a weak shortlist, the failure mode we withdrew a Mars claim over. It also survives 6 of 6 sampled realisations of the archive's own error model (of 100 published; Site01 only). Three things bound it: the three declared axes are one gradient (mean slope and 3×3 relief satisfy relief = 2·d·tan(slope)·ρ, with ρ provably in [1, √2] and checked against the engine's own functions to 4.4e-16 m); a stress test declared before the run leaves the ordering untouched in 28 of the 30 perturbations; and an earlier version of this comparison did not put both bodies on the same ground at all — now fixed and rerun, the ordering held, the distances moved.

EVIDENCE & REPRODUCIBILITY

Every number has a receipt

970
offline tests across 101 modules, 0 failures

51
scientific-correctness oracles, stable hash

4
frozen artifact hashes, re-asserted after every change

0
credentials needed for the headline Earth data path

Reproduce the headline result

```
pip install numpy rasterio
python -m terratrace.cli study \
    --config config/matched_footprint_v3.yml --source cog
```

Offline once the tiles are cached. The engine re-derives two descriptors through an independent GIS toolchain, and validates its reprojection against closed-form geodesy rather than against a second wrapper around the same library.

What keeps it honest

- A guard computes every published count from the source and fails the build if prose drifts.
- A frozen claim ledger maps each claim to the evidence that would falsify it.
- The site's links, fragments and markup are checked before deployment.

Repository

Named first, not buried

Open, and stated plainly

- The lunar work is exploratory: six nearby sites are not six independent tests.
- The error-ensemble check uses 6 of the 100 published realisations.
- No external reviewer has examined the independent reproduction.
- No demo video; the shot-by-shot script is in the repository.

The honest ask

The engine is prior art, built weeks before the event. What a team adds from here is the part that counts: a second terrain family it has never seen, an independent re-derivation by a different GIS toolchain, or a validation set drawn by someone with planetary field experience. If you would rather find the place where this quietly overclaims than add a feature, that is the most useful thing you could do to it.

terratrace.fit

`github.com/pal-123456789/terratrace`

Method write-up, frozen protocols, claim ledger, validation report and limitations — all public.